Which Pass Rate Answers Which Question?

AP Statistics · Unit 2 · Topic 2.2 · B: 2026-10-16 · E: 2026-10-30 · TEACHER KEY

CED TETHER

- **Skill 2.C** — Calculate summary statistics, relative positions of points within a distribution, correlation, and predicted response.
- **Skill 2.D** — Compare distributions or relative positions of points within a distribution.
- **Enduring Understanding UNC-1** — Graphical representations and statistics allow us to identify and represent key features of data.
- **LO UNC-1.Q** — Calculate statistics for two categorical variables. [Skill 2.C]
- **EK UNC-1.Q.1** — The marginal relative frequencies are the row and column totals in a two-way table divided by the total for the entire table.
- **EK UNC-1.Q.2** — A conditional relative frequency is a relative frequency for a specific part of the contingency table (e.g., cell frequencies in a row divided by the total for that row).
- **LO UNC-1.R** — Compare statistics for two categorical variables. [Skill 2.D]
- **EK UNC-1.R.1** — Summary statistics for two categorical variables can be used to compare distributions and/or determine if variables are associated.

Essential question: How do the denominator and study design limit what a percentage can say?

DOK-3 skill: 4.B · **FRQ pattern:** `marginal-vs-conditional-association-is-not-cause`

DOK rationale: Part (c) maps competing percentages to different questions, selects the direct conditional comparison for association, diagnoses a mixed-denominator gap, and bounds a causal claim from self-selected groups.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) **DOK:** 1 → 3 **Minutes:** 45

Teacher does

- Board slide up at the bell. Ask what population sits in each denominator without saying either rate is wrong.
- Begin the video follow-along at minute 5.
- Return to the table at minute 33. Circulate and require students to compare like-conditioned rates.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Name the group described by each first-take percentage.
- Complete the video follow-along about joint, marginal, and conditional relative frequencies.
- Finish (a)–(c), revise the first take in the exit line, and turn in the sheet.

Questions to ask

- Who is included in the denominator 300? Who is included in the denominator 75?
- Which two denominators create a direct users-versus-nonusers comparison?

- Where are tutoring users already included inside the overall 75 percent?
- Who decided whether each student used tutoring, and why does that matter for raises?

Adult role
Solo teacher. Circulate 5 pairs. Require every rate to be stated with its conditioning group and separate evidence of association from evidence of causation.

[WARN] WATCH FOR

- Using the overall 75 percent as though it were the nonuser pass rate.
- Reporting 13 percent instead of 13 percentage points and missing the mixed denominators.
- Treating a conditional-rate difference from self-selected groups as proof of a tutoring effect.

SCORING PART (c) — E / P / I

Expected elements

- **maps-rates-to-questions** (required) — Explains that 75 percent is overall while 88 percent is conditioned on tutoring-center use
- **uses-two-conditional-rates** (required) — Decides association using 88.0 percent for users versus about 70.7 percent for nonusers
- **diagnoses-thirteen-point-gap** (required) — Explains that 13 points compares a subgroup with the overall marginal rate and gives the direct gap of about 17.3 points
- **rejects-causal-wording** (required) — Rejects raises because tutoring was self-selected and names a plausible alternative explanation

E	Maps both reported rates to their questions, supports association with both conditional rates, corrects the mixed-denominator 13-point comparison, and explains why self-selection blocks causation.
P	Correctly distinguishes marginal and conditional rates and identifies association but omits the direct gap, the mixed-denominator diagnosis, or a concrete self-selection explanation.
I	Says one of 75 or 88 must be wrong, uses the marginal rate as the nonuser rate, or concludes that the observed table proves tutoring raised pass rates.

Common mistakes

- Treating 75 percent as the pass rate among nonusers
- Subtracting a conditional rate and marginal rate as though they were the two comparison groups
- Calling the 17.3-point association causal because the users performed better

[STAR] TODAY'S PROBLEM — annotated

Suppose a school records exam outcomes for 300 students. Students chose for themselves whether to use the tutoring center.

Mara: *The pass rate is 75%.*

Noah: *The pass rate is 88%.*

Elena: *Tutoring raises the pass rate by 13 percentage points because* $88\% - 75\% = 13$ *percentage points.*

Exam outcomes for 300 students

Center?	Passed	Did not pass	Total
Used	66	9	75
Did not	159	66	225
Total	225	75	300

FIRST TAKE (5 min)

Can 75% and 88% both be correct pass rates? State what group you think each percentage describes.
Key: Not graded. Expect some students to believe two different pass rates cannot both be correct until their denominators are named.

Rules callout "NAME THE DENOMINATOR, THEN THE CLAIM (keep on the page)" is printed on the student sheet.

DOK 1 **(a)** Identify 75% and 88% as marginal or conditional relative frequencies. For the conditional frequency, name the group on which it is conditioned.

Key: The 75% is the marginal relative frequency of passing among all 300 students. The 88% is the conditional relative frequency of passing among students who used the tutoring center; the condition is used the center.

DOK 2 **(b)** Verify both percentages from the table. Then compute the pass rate among students who did not use the center and the percentage-point gap between the two user groups.

Key: Overall pass rate: $225/300 = 75\%$. Tutoring-user pass rate: $66/75 = 88\%$. Nonuser pass rate: $159/225 = 70.666\ldots\% \approx 70.7\%$. The direct user-minus-nonuser gap is $88.0\% - 70.7\% \approx 17.3$ percentage points, not 13 points.

DOK 3 **(c)** Explain which question Mara's 75% answers and which question Noah's 88% answers. Decide whether tutoring use and exam outcome appear associated, citing the two conditional pass rates that settle it. Then evaluate Elena's 13-point claim: state what two rates she compared, what comparison directly measures the group gap, and why the word *raises* is not warranted when students chose tutoring.

Key: Mara answers What proportion of all 300 students passed? Noah answers What proportion of tutoring-center users passed? Tutoring use and outcome appear associated because the conditional pass rates differ: 88.0% among users versus about 70.7% among nonusers. Elena's 13 points subtracts the user conditional rate from the overall marginal rate, which includes both groups; it is not the direct group comparison. The direct gap is about 17.3 points. Even that association does not establish that tutoring raised pass rates because students self-selected; motivation, prior preparation, or another difference could affect both tutoring use and passing.

EXIT

One thing the video changed about my first take: ____________________